KILNDECK · RISK REGISTER

KilnDeck risk register and hazard analysis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-28	Amish Chadha	First issue (kit 1.8 upgrade) with preliminary hazard analysis and first-pass design FMEA (P3)

KilnDeck risk register and hazard analysis

The top risks are a kiln wall giving way under a track pad, a fall or a burn on a deck that carries a person over a burning kiln, and a deck blown along the kiln in a storm; the design controls each on paper, and each needs the partner kiln survey or a TRL 4 proof load before anyone stands on it. Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high).

> **Safety:** KilnDeck carries a person over a burning kiln and has moving machinery. Nothing in this register authorises use. The guardrail, deck and track must be proof-loaded and the kiln walls surveyed by a competent person before anyone stands on the deck over a fire (TRL 4, on hold).

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	No co-design partner	Neither candidate approached yet	No survey, no firemen's view, no trial	3	5	Amish	Approach is Amish's decision (KND-DDR-001, D8)
P2	Wall bearing limit below 0.15 MPa	Old, hot or cracked kiln walls	Larger pads or a different support; PR-008 not met	2	4	Structural engineer	Survey the partner kiln before TRL 4
P3	Firemen reject a deck over the fire	Radiant heat, unfamiliar, slower at first	No use	3	5	Co-design partner	Interviews first; option B kept on file
P4	Floor too hot by day	Sun on the tread sheet	PR-007 not met; burns through thin soles	4	3	Design lead	Shade canopy with a wind check at TRL 4 (KND-DDR-002, A3)
P5	Piece masses above the team guide	40 kg limit decided before the guide was checked	Strain on the carriers	4	3	Amish	D14
P6	Prices higher than estimated	Rigging and bearings estimated, not quoted	Cost above USD 2,787	3	2	Design lead	Quotes at TRL 4; target margin USD 713

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Fall from the deck into the kiln	Slip, lean on the rail, gate left open	Fatal burns	5	2	1.1 m top rail, knee rail and toe board all round; inward-opening self-closing gate; guardrail checked at 1.5 kN (S5)	1 to 2 after proof load
H2	Deck runs off the track or drops off it	Pushed past the end, wheel lifts	Deck and person fall onto the crust	5	2	End stops (S16), buffers, derailment guards	1
H3	Deck blown along the kiln	Pre-monsoon storm	Deck hits the stops or leaves the track	5	3	Storm clamps and tie-downs (D12, S20); working limit about 7 m/s (D11)	2; anchors to confirm (S21)
H4	Wall gives way under a pad	Overload of an old wall	Track and deck drop	5	2	Pads at 0.15 MPa against an interim 0.20 MPa; survey	To confirm (S9)
H5	Burns from hot surfaces	Floor at 74 °C in full sun, rails in the sun, parts near the crust	Burns to feet and hands	3	4	Insulated floor, grip sleeves, heat-resistant footwear	3 until a canopy or shade
H6	Crushed fingers	Chains, sprockets, rotor, hand wheel spokes	Finger injury	3	2	Sheet guards and a sealed chain case; hopper grille; low crank force (S15)	1
H7	Deck moves while boarding or feeding	Lock pin out	Fall or trapped foot	4	2	Lock pin; gate at the hand wheel end	2
H8	Heat stress and smoke	Working over the fire in summer	Illness	4	4	Not controlled by the deck; shade, water and rest breaks stay necessary	4
H9	Strain carrying pieces up the kiln	40 kg two-person carries	Back injury	3	3	Two people per piece; D14 proposes 33 kg	3 until D14
H10	Fall during assembly	Working at the wall edge	Fall into the trench	5	2	Assemble over the cold zone only (KND-BLD-001)	2
H11	Fire or hot gas flashback into the hopper or spout	Spout held over an open feed hole; hot gas and sparks rise from the fire	Burns to the fireman, coal in the hopper ignites	4	2	Grille and a swing spout lifted clear between holes; hopper filled only from the outer end; to assess with the partner (added after the independent review of 2026-10-29)	3

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Guardrail splice and bearer end plate	Hold a push on the top rail	Bolts loosen	Rail gives under a push	Vibration, heat cycles	5	2	3	30	Nyloc or double nuts; check torque each week (TRL 4 instructions)
F2	Track wheel	Carry the deck	Bearing seizes or wheel cracks	Deck sticks or drops	Dust, heat, overload	4	3	3	36	650 kg rating, shield the bearings, inspect daily
F3	Track pad and wall	Spread load to the wall	Wall top crumbles	Track settles or drops	Weak wall	5	2	4	40	Survey; inspect the wall at each relay
F4	Lock pin	Hold the deck parked	Pin not engaged or sheared	Deck rolls	Operator error, wind	4	3	2	24	Spring plunger; storm clamps for weather
F5	Storm clamp and tie-down	Hold the deck in a storm	Anchor pulls out	Deck moves along the kiln	Soft soil	5	2	4	40	Anchor pull test at the partner kiln (S21)
F6	Floor panel	Carry and insulate the fireman	Board crushed or wet	Hot spots, soft floor	Heel load, rain	3	3	3	27	Board strength to confirm; drain holes
F7	Metering rotor	Dose fuel	Jams with wet coal	No fuel, or a forced crank	Moisture, oversize lumps	2	4	2	16	44 mm grille, crank turns back; TRL 4 trial (PR-005)
F8	End stop	Stop the deck	Bolts shear	Deck runs off	Impact at speed	5	1	3	15	4.0 kN against 81 kN (S16)
F9	Travel drive chain	Drive both ends together	Chain breaks or jumps	Deck skews, binds	Wear, dust	3	2	3	18	Guarded chains, tension check
F10	Gate	Close the end	Fails to close	Open edge	Spring weak, bent leaf	5	2	2	20	Self-closing check at each shift (TRL 4)